

Karthik Sudhakar Kashyap

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EDUCATION

University of Florida

Master of Science in Computer Science

Distributed Operating System, Advanced Data Structures, Large Language Models

Gainesville, FL

Aug. 2024 – May 2026

Dayananda Sagar College of Engineering

Bachelor of Engineering

Bangalore, India

Aug. 2016 – Aug 2020

EXPERIENCE

Software Engineer Intern

May 2025 – August 2025

Juniper Networks

Sunnyvale, California

- Accelerated feature testing cycles by **77%** by building a framework to **validate telemetry streams** over **gRPC & UDP** against schema definitions.
- Scaled** validation capacity from 500 paths and 200k updates/hr to **8k+ paths and 5M+ updates/hr** by building a multi-threaded validation engine.
- Increased test coverage by 45%** by automating telemetry schema extraction with an **ANTLR**-based Protobuf parser.
- Laid the foundation for a **Golang**-based converter to map legacy **Protobuf** files to OpenConfig **YANG** models.

Software Development Engineer - 2

April 2022 – August 2024

Zetwerk

Bangalore, India

- Led a team** to develop a logistics app with **Google Maps** integration, tracking **20,000+ containers and trucks**
- Built a service to **analyze MongoDB logs** and **recommend indexes**, optimizing and reducing costs by **\$10,000/year**.
- Built a plugin to redistribute queries in a **distributed MongoDB** deployment, reducing load on primary node by **65%**.
- Led the **recruitment** process for **DevOps & SDE** roles, **conducting interviews** for 30+ candidates

Software Development Engineer - 1

December 2020 – April 2022

Zetwerk

Bangalore, India

- Engineered a schema-driven low-code form generator, powering **\$2M/monthly inquiries**, reducing dev time by 70%
- Implemented a multi-threaded web scraper to automatically source job leads that converted into **\$3M+** in new contracts.
- Built **Terraform** configurations to automate **infrastructure provisioning**, cutting infra setup time by **80%**

Full-Stack Development Intern

March 2020 – September 2020

Netgear

Bangalore, India

- Developed a **Django-based observability platform** (MySQL, Redis, Celery) for real-time wireless device monitoring, reducing manual validation effort by 60%.
- Implemented **Server-Sent Events (SSE)** for real-time alerting, enabling faster failure detection and resolution.
- Developed automated regression testing using **Robot Framework** to fetch health statistics & generate crash reports

Software Engineering Intern

February 2020 – August 2020

Resolute AI

- Replaced server polling with **WebSockets** using **Django Channels**, improving user experience.
- Developed backend **REST APIs** for an AI application and set up async processing with **Celery** and **RabbitMQ**
- Integrated **AWS Kinesis** for real-time video stream processing in a pilot project, **reducing latency by 30%**
- Setup **CI pipeline** on **GitHub Actions** with **Unit Tests** and **SonarQube**, increasing code coverage from 15% to **78%**

PROJECTS

Syncable | Founder | *MongoDB, Angular, Express, Node.js, Docker, AWS*

September 2023 – Present

- Spearheaded the development of a **SaaS** platform for event and studio management with integrated **ticketing**, processing **\$18,000+** in transactions and scaling to **300+ active users**
- Implemented **secure multi-tenancy**, ensuring strict financial data separation currently supporting **8 organizations**
- Built a Dockerized CI/CD pipeline with **AWS ECR**, cutting deployment time by **50%** on **AWS EC2**.
- 3rd party API integrations: **SendGrid, Razorpay, NewRelic, and Logz.io** to enhance functionality and **observability**

TECHNICAL SKILLS

Languages

JavaScript, Python, TypeScript, Golang, C/C++, Java, HTML/CSS, SQL

Databases

Postgres, MySQL, MongoDB, DynamoDB, Kafka, Redis, RabbitMQ, ElasticSearch, Prometheus

Frameworks

Node.js, Express.js, Django, Flask, Angular, React, Gin, Fast API

DevOps/Cloud

Git, Docker, Terraform, Jenkins, Linux/Unix, Nginx, Kubernetes, AWS, Grafana, System Design